Text 1

Nearly 2,000 years ago, as the Romans began to pull out of Scotland, they left behind a curious treasure: 10 tons of nails, nearly a million of the things. The nail hoard was discovered in 1960 in a four-metre-deep pit covered by two metres of gravel.

Why had the Romans buried a million nails? The likely explanation is that the withdrawal was rushed, and they didn't want the local Caledonians getting their hands on 10 tons of weapon-grade iron. The Romans buried the nails so deep that they would not be discovered for almost two millennia.

Later civilisations would value the skilled blacksmith's labour in a nail even more than the raw material. As Roma Agrawal explains in her new delightful book Nuts and Bolts, early 17th-century Virginians would sometimes burn down their homes if they were planning to relocate. This was an attempt to recover the valuable nails, which could be reused after sifting the ashes. The idea that one might burn down an entire house just to reclaim the nails underlines how scarce, costly and valuable the simple-seeming technology was.

The price of nails fell by 90% between the late 1700s and mid-1900s, as economist Daniel Sichel points out in a research paper. According to Sichel, although the falling price of nails was driven partly by cheaper iron and cheaper energy, most of the credit goes to nail manufacturers who simply found more efficient ways to turn steel into nails.

Nails themselves have changed over the years, but Sichel studied them because they haven't changed much. Roman lamps and Roman chariots are very different from LED strips and sports cars, but Roman nails are still clearly nails. It would be absurd to try to track the changing price of sports cars since 1695, but to ask the same question of nails makes perfect sense.

I make no apology for being obsessed by a particular feature of everyday objects: their price. I am an economist, after all. After writing two books about the history of inventions, one thing I've learnt is that while it is the enchantingly sophisticated technologies that get all the hype, it's the cheap technologies that change the world.

The Gutenberg printing press transformed civilisation not by changing the nature of writing but by changing its cost—and it would have achieved little without a parallel collapse in the price of surfaces to write on, thanks to an often-overlooked technology called paper. Solar panels had a few niche uses until they became cheap; now they are transforming the global energy system.

Text 2

Parenting tips obtained from hunter-gatherers in Africa may be the key to bringing up more contented children, researchers have suggested. The idea is based on studies of communities such as the Kung of Botswana, where each child is cared for by many adults. Kung children as young as four will help to look after younger ones and "baby-wearing", in which infants are carried in slings, is considered the norm.

According to Dr Nikhil Chaudhary, an evolutionary anthropologist at Cambridge University, these practices, known as alloparenting, could lead to less anxiety for children and parents.

Dr Annie Swanepoel, a child psychiatrist, believes that there are ways to incorporate them into western life. In Germany, one scheme has paired an old people's home with a nursery. The residents help to look after the children, an arrangement akin to alloparenting. Another measure could be encouraging friendships between children in different school years to mirror the unsupervised mixed-age playgroups in hunter-gatherer communities.

In a paper published in the Journal of Child Psychology and Psychiatry, researchers said that the western nuclear family was a recent invention which broke with evolutionary history. This abrupt shift to an "intensive mothering narrative", which suggests that mothers should manage childcare alone, was likely to have been harmful. "Such narratives can lead to maternal exhaustion and have dangerous consequences," they wrote.

By contrast, in hunter-gatherer societies adults other than the parents can provide almost half of a child's care. One previous study looked at the Efe people of the Democratic Republic of Congo. It found that infants had an average of 14 alloparents a day by the time they were 18 weeks old and were passed between caregivers eight times an hour.

Chaudhary said that parents now had less childcare support from family and social networks than during most of humans' evolutionary history, but introducing additional caregivers could reduce stress and maternal depression, which could have a "knock-on" benefit to a child's wellbeing. An infant born to a hunter-gatherer society could have more than ten caregivers—this contrasts starkly to nursery settings in the UK where regulations call for a ratio of one carer to four children aged two to three.

While hunter-gatherer children learnt from observation and imitation in mixed-age playgroups, researchers said that western "instructive teaching", where pupils are asked to sit still, may contribute to conditions such as attention deficit hyperactivity disorder. Chaudhary said that Britain should explore the possibility that older siblings helping their parents "might also enhance their own social development."

Text 3

A Polish digital artist who uses classical painting styles to create dreamy fantasy landscapes, Greg Rutkowski has made illustrations for games such as Dungeons & Dragons and Magic: The Gathering. And he's become a sudden hit in the new world of text-to-image AI generation.

His distinctive style is now one of the most commonly used prompts in the new open-source AI art generator Stable Diffusion, which was launched late last month. The tool, along with other popular image-generation AI models, allows anyone to create impressive images based on text prompts. For example, type in "Wizard with sword and a glowing orb of magic fire fights a fierce dragon Greg Rutkowski," and the system will produce something that looks not a million miles away from works in Rutkowski's style.

But these open-source programs are built by scraping images from the internet, often without permission and proper attribution to artists. As a result, they are raising tricky questions about ethics and copyright. And artists like Rutkowski have had enough.

According to the website Lexica, which tracks over 10 million images and prompts generated by Stable Diffusion, Rutkowski's name has been used as a prompt around 93,000 times. Some of the world's most famous artists, such as Michelangelo, Pablo Picasso, and Leonardo da Vinci, brought up around 2,000 prompts each or less. Rutkowski's name also features as a prompt thousands of times in the Discord of another text-to-image generator, Midjourney. Rutkowski was initially surprised but thought it might be a good way to reach new audiences. Then he tried searching for his name to see if a piece he had worked on had been published. The online search brought back work that had his name attached to it but wasn't his.

"It's been just a month. What about in a year? I probably won't be able to find my work out there because the internet will be flooded with AI art," Rutkowski says. "That's concerning."

Other artists besides Rutkowski have been surprised by the apparent popularity of their work in text-to-image generators—and some are now fighting back.

Karla Ortiz, an illustrator based in San Francisco who found her work in Stable Diffusion's data set, has been raising awareness about the issues around AI art and copyright. Artists say they risk losing income as people start using AI-generated images based on copyrighted material for commercial purposes. But it's also a lot more personal, Ortiz says, arguing that because art is so closely linked to a person, it could raise data protection and privacy problems.

"There is a coalition growing within artist industries to figure out how to tackle or mitigate this," says Ortiz. The group is in its early days of mobilization, which could involve pushing for new policies or regulation. One suggestion is that AI models could be trained on images in the public domain, and AI companies could forge partnerships with museums and artists, Ortiz says.

Text 4

The miracle of the Chesapeake Bay lies not in its depths, but in the complexity of its natural construction, the interaction of fresh and saline waters, and the mix of land and water. The shallows provide homes for hundreds of species while storing floodwaters, filtering pollutants from water, and protecting nearby communities from potentially destructive storm surges.

All this was put at great risk late last month, when the U.S. Supreme Court issued a ruling in an Idaho case that provides the U.S. Environmental Protection Agency (EPA) far less authority to regulate wetlands and waterways. Specifically, a 5-4 majority decided that wetlands protected by the EPA under its Clean Water Act authority must have a "continuous surface connection" to bodies of water. This narrowing of the regulatory scope was a victory for builders, mining operators and other commercial interests often at odds with environmental rules. And it carries "significant repercussions for water quality and flood control throughout the United States," as Justice Brett Kavanaugh observed.

In Maryland, the good news is that there are many state laws in place that provide wetlands protections. But that's a very shortsighted view, particularly when it comes to the Chesapeake Bay. The reality is that water, and the pollutants that so often come with it, don't respect state boundaries. The Chesapeake draws from a 64,000-square-mile watershed that extends into Virginia, Pennsylvania, New York, West Virginia, the District of Columbia and Delaware. Will those jurisdictions extend the same protections now denied under Sackett v. EPA? Perhaps some, but all? That seems unlikely.

It is too easy, and misleading, to see such court rulings as merely standing up for the rights of land owners when the consequences can be so dire for their neighbors. And it's a reminder that the EPA's involvement in the Chesapeake Bay Program has long been crucial as the means to transcend the influence of deep-pocketed special interests in neighboring states. Pennsylvania farmers, to use one telling example, aren't thinking about next year's blue crab harvest in Maryland when they decide whether to spread animal waste on their fields, yet the runoff into nearby creeks can have enormous impact downstream.

And so we would call on state lawmakers from Richmond to Albany to consider reviewing their own wetlands protections and see for themselves the enormous stakes involved. We can offer them a visit to Blackwater National Wildlife Refuge in Dorchester County where bald eagles fly over tidal marshes so shallow you could not paddle a boat across them but teeming with aquatic life. It's worth the scenic drive.